EXORA

HOSPITALITY INTELLIGENCE

Complete Build Project Specification

Dynamic Room-Price Recommendation Engine + Unified Operations & Financial Intelligence

PROJECT PURPOSE

Build an independent, explainable pricing recommendation engine using a shared hotel-revenue data foundation. Start from the publicly documented functional baseline of ChoiceADVANTAGE and Wyndham RevIQ, then add the extra signals needed to make recommendations more contextual, measurable and useful. Exora will not copy proprietary algorithms or depend on restricted outputs.

Prepared for: Exora project team and motel prospect

Properties in discovery: Travelodge Fargo West Acres (~70 rooms) and Quality Suites Fargo West Acres (~50 rooms)

Version 1.0 | 04 November 2026 | Status: Proposed build specification, subject to client data and vendor access validation

Confidential working document. Product capabilities and data access statements are separated into client-reported facts, public documentation and items requiring confirmation.

Document map

Section	Contents
1	Executive summary and product thesis
2	Client context and current systems
3	What is documented about the incumbent tools
4	Target product scope and operating model
5	Data strategy and variable catalogue
6	Metrics and definitions
7	Pricing recommendation methodology
8	System architecture and flows
9	Data model and ingestion contracts
10	Dashboard, reporting and AI experience
11	Security, governance and contractual controls
12	Testing, backtesting and success measures
13	Implementation roadmap and acceptance criteria
14	Risks, assumptions and discovery checklist
15	References and source notes

READ THIS FIRST

This document is a build specification, not proof that every API, export field, licensing term or client workflow has been confirmed. The first project gate is to inspect both properties' actual CSV exports and obtain the applicable franchise/vendor permissions.

1. Executive summary and product thesis

The prospect operates two franchise-affiliated motels in Fargo, North Dakota: Travelodge Fargo West Acres, associated with Wyndham Hotels & Resorts, and Quality Suites, associated with Choice Hotels. The working room-count assumptions are approximately 70 rooms and 50 rooms respectively (about 120 total); confirm exact sellable inventory and room classes with the client.

The properties already use hotel reservation/property systems and pricing or revenue-management functionality. The project is not to replace their franchise systems. Exora should sit above permitted source systems as a decision-support and analytics layer.

The central build is an independent pricing recommendation engine. It will use the business signals that are publicly documented in the incumbent systems' revenue-management descriptions—especially historical property performance and projected occupancy—and extend that baseline with booking pace, inventory availability, rate history, competitor data, event and weather context, channel economics and property-specific positioning where data supports those features.

The platform should also consolidate daily/night operational and financial reports, because the client described a manual management burden around ADR, cash flow, revenue excluding tax, discrepancies, no-shows, cancellations and the hotel ledger. That reporting capability shares the same data foundation but is a distinct product module from the pricing algorithm.

PRODUCT POSITIONING

Exora Hospitality Intelligence = a unified data platform + a hotel pricing decision engine + operations/financial analytics + an explainable management interface. The pricing engine is the core differentiator; AI is used to interpret documents and explain validated calculations, not to invent financial numbers or independently control prices.

1.1 Business outcomes sought

- Recommend a room rate for each property, stay date and supported room class, with an explanation of why the rate is recommended.
- Forecast demand and occupancy early enough for managers to act on future dates, not just summarize yesterday.
- Compare each property against a relevant, normalized competitive set rather than comparing unqualified public prices.
- Bring both properties into one view while respecting the different franchise ecosystems and each property's own rules.
- Reduce the manager's manual effort reviewing daily/night reports and surface important exceptions.
- Allow a human manager to approve, edit or reject a recommendation before any production rate change.

2. Client context and current systems

2.1 Properties and franchise relationships

Property	Franchise / brand relationship	Reported system	Room-count assumption	Validation needed
Travelodge Fargo West Acres	Wyndham Hotels & Resorts / Travelodge	SynXis (specific module/version to confirm); client reports revIQ in the pricing workflow	~70	Confirm property-system module, revIQ entitlement/version, rate push path, room classes, exact inventory.
Quality Suites Fargo West Acres	Choice Hotels International / Quality Suites	ChoiceADVANTAGE	~50	Confirm active ChoiceADVANTAGE modules, revenue-management capability enabled, export fields and any separate tool.

Client-reported context: bookings arrive from multiple booking platforms; the properties already have franchise software; daily/night reports are produced; CSV exports are available; room prices may be reconsidered multiple times on busy days and less frequently on quieter days; the client stated there are no franchise restrictions on setting prices. That last statement must be distinguished from permission to integrate third-party software or automate rate publishing.

2.2 Current-system visual: end-to-end workflow

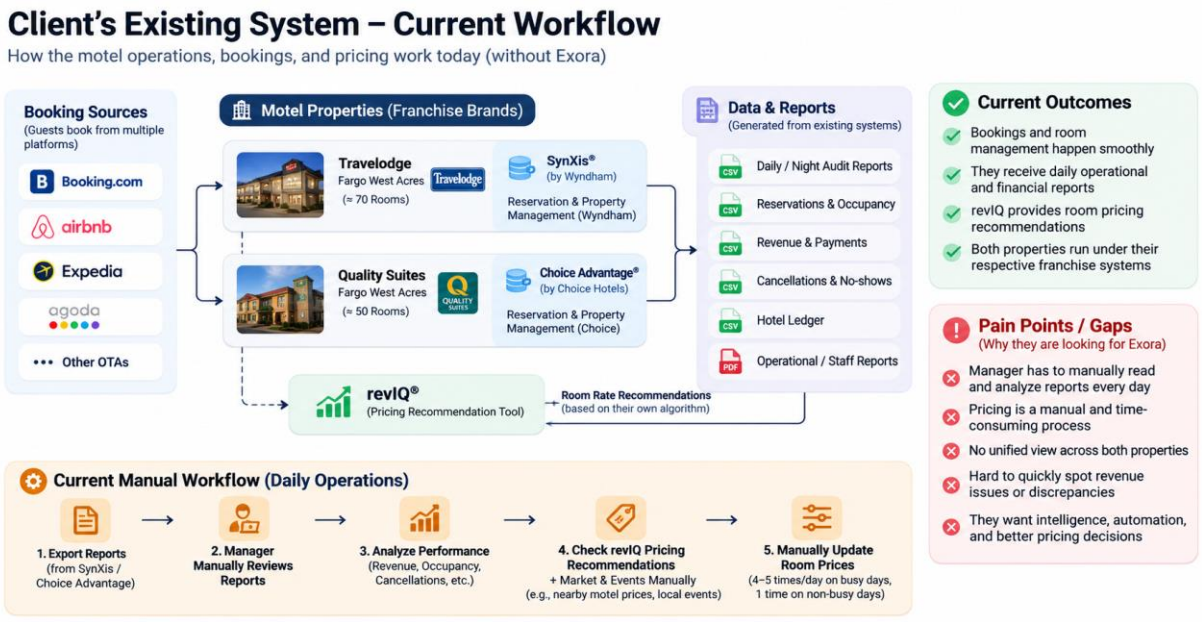


Figure 1. Client-provided/generated illustration of the current operating workflow. Treat system details as a discovery illustration, not a verified integration map.

2.3 Where revIQ participates

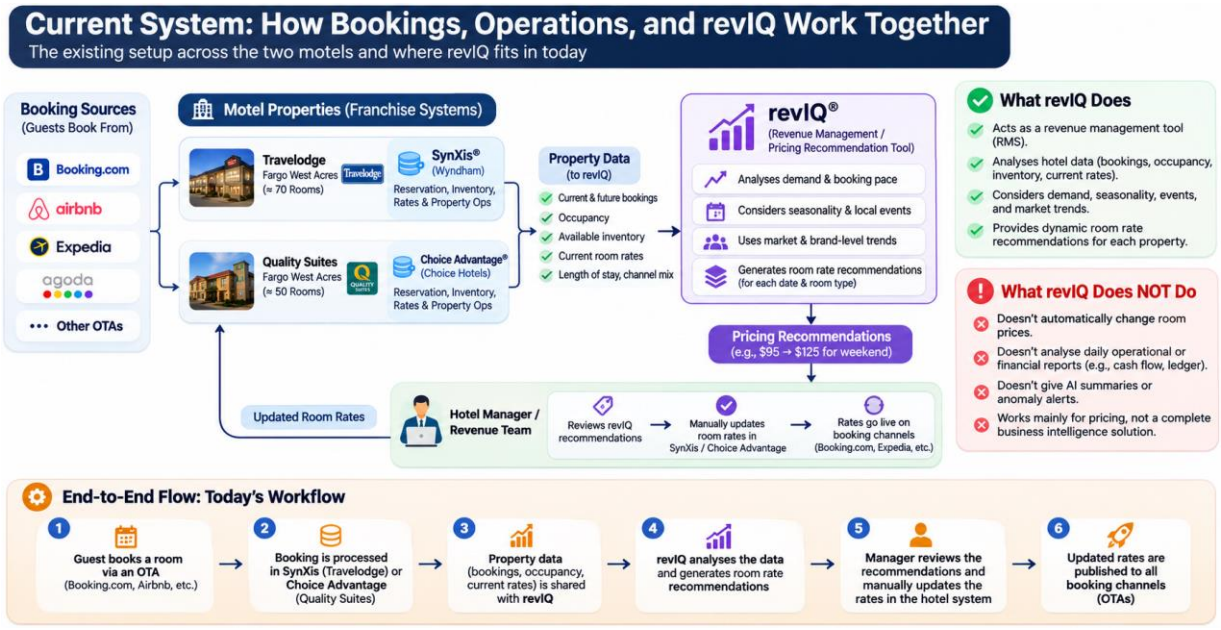


Figure 2. Illustration of revIQ as a revenue-management recommendation layer between property data and the manager's rate-update workflow.

Conceptually, revIQ is a pricing/revenue-management layer, not the franchise itself and not the same thing as the property-management or reservation system. It uses hotel and market context to produce rate decisions or recommendations. Depending on the product configuration and supported interface, price decisions may be transmitted to connected reservation systems; the actual Travelodge property workflow must be verified.

2.4 Target-state visual: what the client wants from Exora



Figure 3. Target-state concept: preserve the existing franchise systems and add Exora's shared intelligence layer.

2.5 Existing capability vs Exora responsibility

Existing component	Role today / public description	Exora approach

Existing component	Role today / public description	Exora approach
SynXis / Wyndham stack	Reservation/property data and distribution functions; SynXis Property Hub exposes reservation retrieval APIs in its developer documentation. [R3]	Use client-authorized CSV exports first; investigate authorized API access after vendor/property onboarding.
ChoiceADVANTAGE	Choice describes it as a property/yield management system for room inventory, rates and reservations; its rate tool suggests rates based on past property performance and projected occupancy. [R2]	Use exported source records to build a cross-property data view; do not assume the client has API access.
Wyndham RevIQ	A Wyndham/IDeaS revenue-management capability. The 2025 La Quinta RevIQ supplement documents forecasts, pricing optimizations, events/seasons, floors/ceilings and competitive sets for that franchise context. [R1]	Use as a benchmark to understand the workflow only. Build Exora from independently authorized source data and licensed market data.
Daily/night reports	Staff and hotel systems produce reports; client says managers manually review them.	Automate deterministic KPI calculations, exception rules, and AI-supported narrative summaries.
Booking channels / OTAs	Reservations arrive through more than one channel.	Preserve channel/source and channel cost where available; do not assume all channels are directly queryable.

3. What is documented about the incumbent revenue tools

IMPORTANT DISTINCTION

Vendor documentation describes product functions, not the complete proprietary feature vector, weights, coefficients, or optimization algorithm. Exora can adopt a compatible business-variable framework, but cannot claim to know or reproduce the private internal logic of ChoiceADVANTAGE or RevIQ from public descriptions.

3.1 ChoiceADVANTAGE — publicly described baseline

Choice Hotels' 2025 Form 10-K says ChoiceADVANTAGE includes a rate and selling management tool that helps franchisees manage rates and inventory to improve RevPAR by optimizing ADR and occupancy. It also describes a revenue-management feature that calculates suggested optimum rates based on each hotel's past performance and projected occupancy. [R2]

- Documented at a high level: property room inventory, rates, reservations, historical property performance and projected occupancy.
- Not disclosed in the public filing: the complete model input list, forecast method, feature weights, price elasticity method, competitor-rate feed details, exact update frequency or API schema for this property.

3.2 Wyndham RevIQ — publicly described functions

A 2025 RevIQ supplement embedded in a La Quinta franchise disclosure document describes RevIQ Standard as providing a daily optimization for base-price decisions and last-room value for the next 365 days, four intra-day optimizations for the next 14 days, a daily 365-day occupancy forecast, non-base room-type pricing offsets, user-configured special events and pricing seasons, seasonal price floors and ceilings, smart alerts, reporting and user-configured competitive sets. The Premium tier adds up to three priced room classes and optimized hotel/room-type overbooking. [R1]

Scope caveat: the source is a 2025 La Quinta franchise supplement, not the Travelodge owner's executed agreement. It is useful to understand published RevIQ features, but the client's own Travelodge agreement, configuration and RevIQ tier must be obtained before treating any term or feature as applicable.

3.3 Baseline variable families Exora should support

Variable family	Why include it in Exora baseline	Public evidence status
Historical property performance	The public Choice description explicitly references past performance; RevIQ supports long-horizon forecasts and price optimization.	Choice explicitly; RevIQ functions imply performance history but exact fields are not disclosed.

Variable family	Why include it in Exora baseline	Public evidence status
Projected occupancy / demand	Fundamental to rate setting and inventory yield.	Choice explicitly; RevIQ explicitly documents 365-day occupancy forecast.
Room inventory / sellable capacity	Needed to understand how many rooms remain to sell.	Choice explicitly manages inventory; RevIQ price/LRV and overbooking functions depend on property inventory context.
Current/base rates and rate classes	Needed to set and compare a property's prices and room types.	Rates are central to both tools; RevIQ explicitly documents base prices and room-type offsets/classes.
Seasonality and special events	Demand varies by day, season and event.	RevIQ supplement explicitly documents user-configured seasons and special events; Choice's public filing does not describe its event inputs.
Competitor set / competitor rates	A selected competitor group provides market context.	RevIQ supplement explicitly documents configured competitive sets; not confirmed in Choice public description.
Price floors / ceilings	Prevent unreasonable or disallowed recommendations.	RevIQ supplement explicitly documents configurable floors/ceilings; Choice public filing does not specify these controls.

PROJECT DECISION

Use the shared business foundation—historical property performance, projected occupancy/demand, room inventory and room rates—as the common baseline. Add the other documented RevIQ capabilities (events, seasons, comp set, floors/ceilings and room classes) where the client's data, configuration and permissions support them. Do not label every variable as "used by both systems" because Choice's public documentation does not reveal that level of detail.

4. Target product scope and operating model

4.1 Product modules

Module	Main job	MVP priority	Boundary
Data ingestion & normalization	Import CSVs and authorized APIs; normalize two franchise data formats.	P0	No direct scraping or unauthorized access.
Pricing data foundation	Create property/date/room-class histories and daily booking snapshots.	P0	Must be reliable before model development.
Pricing recommendation engine	Forecast demand, evaluate candidate prices, enforce guardrails, explain recommendation.	P0	Advisory only until approval and write-back are proven.
Pricing workbench	30-day calendar, recommendations, competitor context, approve/edit/reject.	P0	Manager remains in control.
Performance measurement	Compare forecasts/recommendations with actual outcomes and incumbents.	P0	Use proper holdouts/backtests; do not promise uplift.
Operations & financial intelligence	Daily/night report parsing, ADR/RevPAR, discrepancies, cancellation/no-show and ledger checks.	P1	Separate reporting module sharing same data layer.
AI management assistant	Natural-language questions and concise explanations over validated metrics.	P1	LLM does not perform authoritative arithmetic.
Automated rate publishing	Push approved rates to PMS/CRS where permitted.	P2	Separate technical and contractual approval gate.

4.2 What Exora is not building

- A replacement for SynXis, ChoiceADVANTAGE, franchise CRS, channel manager or guest booking engine.
- A payment gateway, accounting system or general ledger replacement.

- A black-box model that changes live rates without review during the pilot.
- A reverse-engineered clone of RevIQ or ChoiceADVANTAGE.
- A guest-personalized pricing system. Recommendations are based on property/date/room-class market and inventory conditions, not an individual guest's protected or personal characteristics.

5. Data strategy and variable catalogue

The pricing engine is only as good as its historical and current data. The most important requirement is not simply a daily occupancy total. We need to reconstruct what the property knew about each future stay date at multiple points in time. This is what allows us to calculate booking pace and train forecasts without leaking future information into historical tests.

5.1 Required data domains

Domain	Variables / fields	Purpose	Priority
Property & capacity	property_id, brand, address/geo, timezone, total rooms, sellable rooms, opening days, room classes, amenities, property positioning	Separate the properties and understand capacity and product differences.	P0
Reservations	reservation ID, created_at, modified_at, arrival, departure, room nights, room class, status, channel/source, rate plan, cancellation timestamp, no-show flag	Build booking history, lead time, LOS, cancellations and demand.	P0
Daily inventory snapshots	stay_date, room_class, snapshot timestamp, OTB rooms, available rooms, out-of-order rooms, oversell/overbooking status	Reconstruct booking curves and remaining sellable inventory.	P0
Rates & realized revenue	public/current rate, booked rate, nightly rate, room revenue ex-tax, tax, fees, discounts, currency, rate plan	Measure ADR, net/gross revenue and price history.	P0
Actual stay outcomes	occupied rooms, checked-in/check-out, no-show, cancellations, actual paid room nights, refunds/adjustments where available	Measure forecast outcomes and cancellation/no-show patterns.	P0
Competitive rates	competitor ID, source/channel, stay date, room category, occupancy basis, refundable terms, meal/fees/tax treatment, price, availability, collected_at	Build normalized market reference and rate index.	P1
Seasonality & events	weekday, weekend, month, season, holiday, event name/type, venue, distance, dates, estimated attendance/impact, source confidence	Represent demand context not captured by property history.	P1
Weather & disruptions	forecast/alert type, severity, temperature, snowfall/precipitation, wind, valid time, location/distance	Add relevant travel disruption or demand context.	P1
Channel economics	channel, commission %, payment/transaction costs if provided, net room revenue, cancellation terms	Optimize net contribution rather than headline ADR alone.	P1
Pricing controls	min/max rate, max step change, rate parity constraints, excluded dates, room-class relationships, approval policy	Keep recommendations operationally and contractually valid.	P0
Manager feedback	recommendation_id, recommended rate, approved rate, override reason, approver, timestamp	Capture adoption, reasons and outcomes.	P0

5.2 Common baseline variables (must be available)

Variable	Definition	Initial source	How engine uses it
Historical ADR	Average realized room revenue per occupied room night, excluding taxes/fees under an agreed accounting convention.	Historical reservation/nightly revenue CSV	Baseline performance by property, weekday, month, room class and lead time.
Historical occupancy	Occupied room nights divided by sellable room nights.	PMS/night audit and inventory snapshots	Demand pattern and forecast target.
Projected occupancy / OTB occupancy	Rooms already booked for a future stay date divided by sellable inventory.	Future reservations + inventory export	Current booking position before forecasted future pickup.
Sellable inventory	Physical room count minus out-of-order rooms and other unavailable units.	Inventory/housekeeping/maintenance exports	Capacity ceiling and remaining rooms to sell.
Current/base rate	Current public or approved base rate for property/date/room class.	PMS rate export	Anchor and comparison point for recommendations.

Variable	Definition	Initial source	How engine uses it
Historical property performance	ADR, occupancy, RevPAR, room revenue and pickup by comparable periods.	PMS history / daily snapshots	Seasonality, baseline forecast and performance comparison.
Room class / room-type offset	Room category and price difference relative to base class.	PMS room-type mapping	Separate recommendations by sellable room class where data supports.
Day-of-week and season	Calendar features for the stay date.	Derived calendar / client-configured seasons	Capture recurring local patterns.

5.3 Additional variables Exora should add

Additional variable	Why it matters	MVP / later
Booking lead time	Days between booking creation and arrival; shows when demand is materializing.	P0
Booking pickup / pace	Change in on-the-books rooms between saved snapshots for the same future stay date.	P0
Length of stay (LOS)	One-night and multi-night demand can differ; some market feeds compare LOS.	P0
Cancellation pickup / cancellation rate	A gross reservation count can overstate expected occupied rooms.	P0
No-show rate	Useful for realized demand and operational reporting; must be separated from cancellations.	P1
Channel mix	Direct, brand.com, OTA, GDS, phone/walk-in shares can change net yield.	P1
Commission-adjusted net rate	A $110 OTA booking may net less than a $105 direct booking.	P1 when commission data exists
Competitor availability and matched terms	A competitor rate without availability, room/terms equivalence can mislead.	P1
Competitive rate index	Our rate divided by comparable comp-set median, with direction and freshness.	P1
Event intensity and distance	Event size, venue distance, event dates and confidence improve context over a plain event flag.	P1
Weather severity and forecast confidence	Only severe/local weather should affect demand; avoid simplistic temperature pricing rules.	P1
Amenities / room product attributes	Explains why a room class can command a premium or discount.	P1 after mapping
Brand/property positioning	Separates Travelodge and Quality Suites positioning without assuming brand alone determines willingness to pay.	P1
Out-of-order / maintenance rooms	Capacity should reflect rooms that cannot actually be sold.	P0 if records exist
Rate elasticity / price response	Estimates how booking probability changes as price changes.	P2 after sufficient variation and careful validation
Forecast uncertainty	Prevents aggressive recommendations when model confidence is low.	P0
Data freshness / source quality	A stale competitor rate or missing snapshot should lower confidence or block a recommendation.	P0

5.4 Snapshot requirement — non-negotiable

WHY SNAPSHOTS MATTER

A current reservations export tells us what is booked now. It does not tell us what was booked 30, 14 or 7 days before arrival. Without dated snapshots, we cannot reconstruct historical booking pace accurately. Start saving a daily snapshot of future occupancy, room rates and inventory for each property, stay date and room class from the first pilot day.

- Minimum daily snapshot horizon: next 60 days (30-day horizon is enough for the first demo, but 60 days gives more lead-time context).

- Retain raw reservation changes and snapshot timestamps; never overwrite prior snapshots with the latest totals.
- Capture whether each value is on-the-books, forecast, actual, or manually supplied. These must not be mixed.

6. Metrics and definitions

Before building a recommendation model, the client and Exora must agree on the accounting definitions. The same metric can differ between systems when taxes, fees, complimentary rooms, out-of-order rooms or room revenue are treated differently.

Metric	Formula / definition	Implementation note
Sellable rooms	Physical room inventory − out-of-order/unavailable rooms	Use by stay date and room class where possible.
Occupancy %	Occupied room nights ÷ sellable room nights × 100	Specify whether based on actual stays or on-the-books future reservations.
ADR	Room revenue excluding tax ÷ occupied paid room nights	Agree how complimentary rooms, discounts and refunds are treated.
RevPAR	Room revenue excluding tax ÷ available sellable room nights	Equivalent to ADR × occupancy when definitions align.
OTB occupancy	Confirmed on-the-books room nights ÷ sellable room nights	Future bookings currently held; not the final forecast.
Pickup	OTB room nights at current snapshot − OTB room nights at prior snapshot	Compute over 1/7/14-day windows and by days-to-arrival.
Cancellation rate	Cancelled room nights ÷ total booked room nights in a defined cohort	Use a clear cohort definition and include cancellation timing.
No-show rate	No-show room nights ÷ eligible arrival room nights	Keep separate from cancellation rate.
Booking lead time	Arrival date − booking creation date	Can analyze median/quantiles by property and stay date.
Length of stay	Departure date − arrival date	Expand multi-night reservations to nightly records.
Comp-set index	Property rate ÷ comparable competitor median × 100	Only compute on matched dates, room/occupancy basis, taxes and refund terms.
Net ADR	Net room revenue after known distribution/transaction costs ÷ occupied paid room nights	Only calculate when channel cost data is reliable.

7. Pricing recommendation methodology

7.1 Design principle

Do not ask an LLM to “guess the best rate.” Build a deterministic and statistical pricing service that calculates demand, evaluates price scenarios and returns a constrained recommendation. Use AI afterward to explain the evidence in plain language.

7.2 Decision unit

The target decision key is: property × stay_date × room_class × rate_plan/condition (only where needed). For the first MVP, start at property × stay_date × base room class; add additional priced room classes once the source mapping and volume justify it.

7.3 Model pipeline

1. Create a clean daily panel for every property, stay date and room class, including capacity, OTB, actuals, current price and source freshness.
2. Build comparison features: same weekday, same season, same lead-time bucket, prior-year or comparable historical dates, booking pickup and cancellation pattern.
3. Forecast expected final demand/occupancy before pricing. Include uncertainty intervals and baseline benchmarks.

4. Establish market context using normalized competitor rates and availability, plus event/weather indicators where data quality is sufficient.
5. Estimate how expected bookings respond to candidate prices. Do not assume the competitor median is the optimal price or that higher occupancy always means a price increase.
6. Optimize the candidate rate against expected net room revenue subject to property-specific price, inventory, rate-change and franchise constraints.
7. Create explanation codes, confidence, data-quality warnings and a transparent comparison to current/previous rate.
8. Require manager approval in the pilot; record all edits, rejections and override reasons.
9. Measure actual pickup, occupancy, ADR, RevPAR, net room revenue and forecast error after the stay date.

7.4 Recommended technical progression

Stage	Method	What it is good for	Do not claim
Stage 0: baseline	Comparable-date benchmarks + explicit rule/guardrail engine	Fast demo, auditability, data-gap discovery	Do not call it machine learning or claim optimized revenue.
Stage 1: forecasting	Seasonal-naive baseline, regression/gradient-boosted models, pooled property effects	Forecast demand/occupancy where history is sufficient	Do not train separate complex models on tiny property datasets.
Stage 2: price response	Estimate demand response to price using historical variation, controls and carefully designed experiments	Understand likely demand at alternative rates	Do not infer causality from raw correlation alone.
Stage 3: constrained optimization	Candidate price search over expected occupied rooms × net rate, with business constraints	Produce a feasible rate recommendation	Do not maximize ADR alone or ignore channel cost and inventory.
Stage 4: controlled automation	Approved rate-publishing adapter + monitoring + rollback	Reduce manual rate entry once integrations are authorized	Do not enable autonomous write-back before certification and manager sign-off.

7.5 Objective function (conceptual)

OPTIMIZATION TARGET

For each stay date and room class, evaluate feasible candidate rate r and estimate expected sellable rooms Q(r). A simple objective is expected net room revenue = r × Q(r) × (1 − effective channel cost), with taxes/fees excluded from room revenue. Add operational constraints and, where data allows, incremental variable costs. This is a conceptual formulation; actual objective and cost treatment must be approved by the owner.

The engine must account for the fact that remaining rooms are finite. If occupancy is forecast to exceed capacity, additional demand cannot create more room nights; the objective becomes selecting the best rate/mix under capacity and overbooking rules. Overbooking should not be included in the first release unless the client explicitly asks for it and the source data supports cancellations/no-shows accurately.

7.6 Confidence and fallback policy

Condition	Engine behaviour
All critical PMS inputs fresh; snapshots consistent; enough history	Generate normal recommendation with confidence band.
Competitor feed stale or unavailable	Use internal-data model; label market context missing; lower confidence.
Missing inventory or out-of-order data	Block or clearly mark recommendation as non-publishable.
Insufficient historical data for the room class	Use property-level / pooled baseline and conservative guardrails; do not pretend room-class model is trained.
Extreme event or severe weather with no comparable history	Show event/alert as context; do not apply an arbitrary large price multiplier.
Model drift / high forecast error	Fall back to approved baseline rules and notify revenue manager.

8. System architecture and flows

8.1 Target architecture

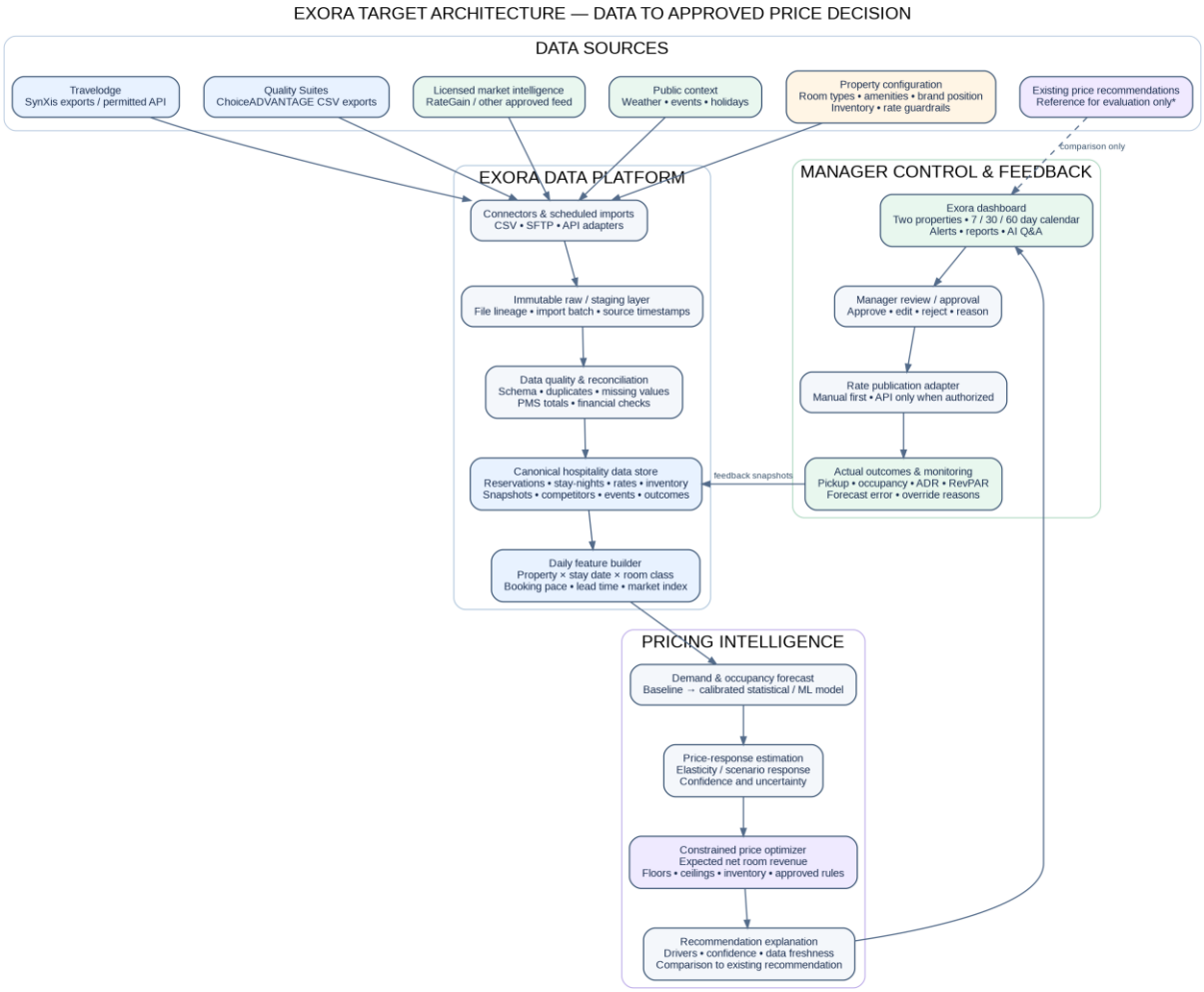


*Figure 4. Proposed Exora technical architecture. *Existing revenue recommendations may be compared only where the client and franchise contract permit; they are not assumed to be a model-training data source.*

8.2 Pricing recommendation workflow

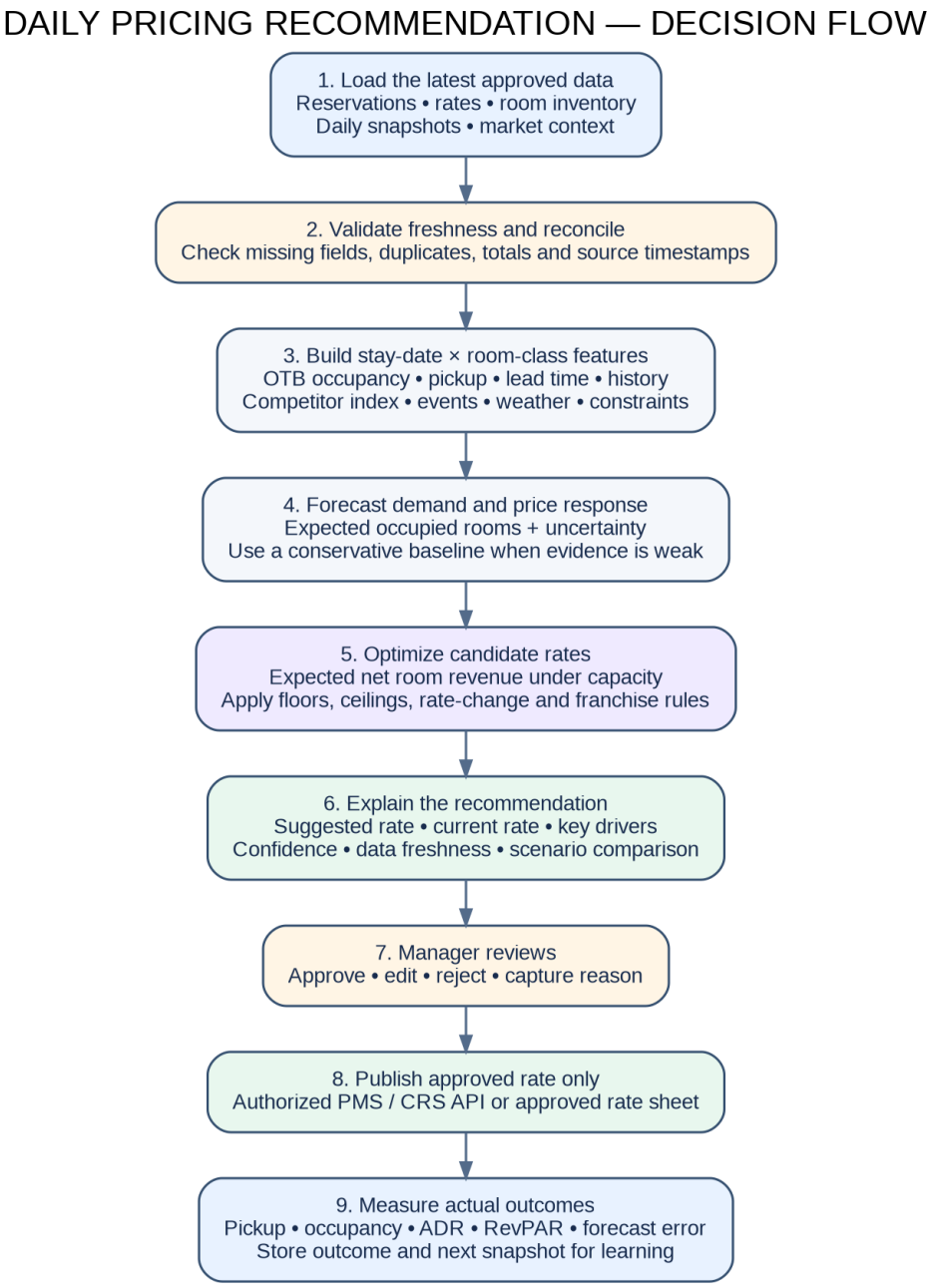


Figure 5. Daily price-decision workflow with validation gates, human approval and post-stay feedback. Invalid or stale data should hold the recommendation rather than silently fall back to an unsafe rate.

8.3 Logical data model

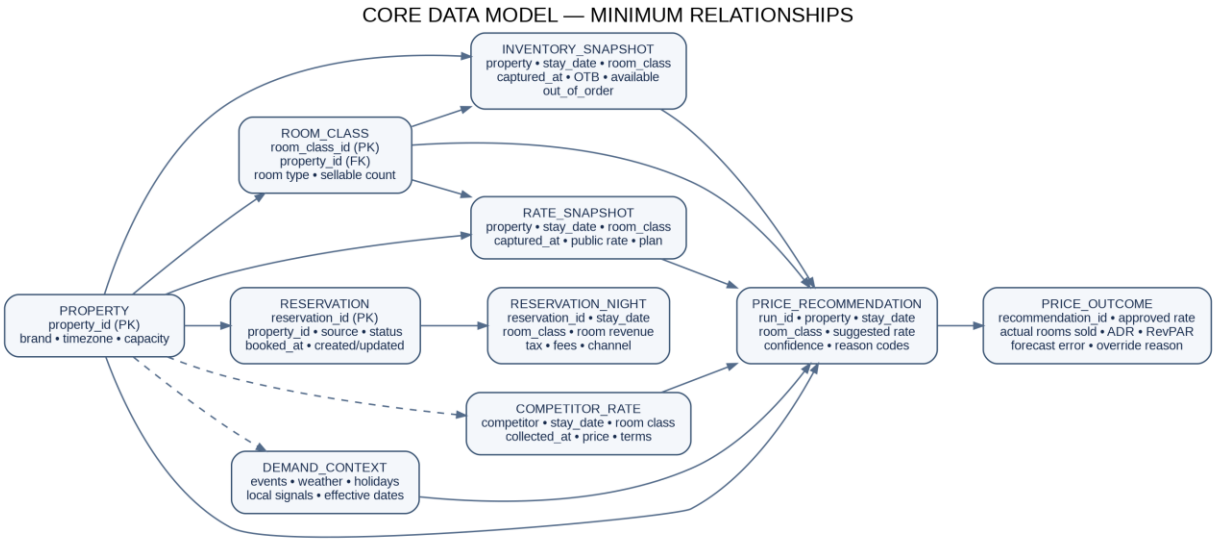


Figure 6. Minimum core entities and their relationships. This is a conceptual model; physical schema is refined after CSV inspection.

8.4 Suggested service boundaries

Service	Responsibility	Suggested implementation
Web dashboard	Property overview, calendar, recommendation details, approvals, analytics, report viewer	Next.js / React / TypeScript
Application API	Auth, properties, ingestion runs, recommendation queries, approval workflow	Fastify or NestJS / TypeScript
Data ingestion workers	Parse CSV/XLSX, normalize source records, deduplicate, validate, load	Python or TypeScript worker; versioned mapping profiles
Pricing service	Feature engineering, demand forecast, candidate-rate evaluation, optimizer, explanations metadata	Python service (FastAPI) for DS libraries; batch jobs first
Database	Transactional domain data, snapshots, model runs, approvals and audit history	PostgreSQL; partition high-volume snapshots if needed
Object storage	Original reports/exports, immutable ingestion artifacts, model artifacts	S3-compatible object storage with encryption
Scheduler / queue	Scheduled imports, daily snapshots, model runs, alerts	BullMQ/Redis or managed scheduler/queue
Observability	Logs, job status, data freshness, model metrics and alerting	Structured logs + metrics + error tracking

This is a recommended implementation shape, not a mandatory stack commitment. The pricing engine should be isolated from the UI and franchise adapters so the team can replace market-data providers or add properties without rewriting the model.

9. Data model and ingestion contracts

9.1 Canonical entities

Entity	Key fields	Notes
property	property_id, franchise, brand, name, timezone, currency, address, room_count	Use stable internal IDs; never rely on names as keys.
room_class	room_class_id, property_id, source_room_type_code, canonical_class, capacity, base_offset	Maintain source-to-canonical mapping and effective dates.
reservation	reservation_id, source_system, property_id, created_at, updated_at, arrival, departure, status, channel	Store source IDs and normalized status mapping.

Entity	Key fields	Notes
reservation_night	reservation_id, stay_date, room_class_id, rate, room_revenue, tax, fees, currency	One row per reservation-night for analysis.
inventory_snapshot	property_id, stay_date, room_class_id, captured_at, physical, OTB, available, out_of_order	Append snapshots; do not overwrite.
rate_snapshot	property_id, stay_date, room_class_id, captured_at, base_rate, selling_rate, rate_plan	Preserve effective date and collected timestamp.
competitor_rate_observation	competitor_id, property_id, stay_date, room_class, source, price, terms, available, collected_at	Store comparable terms, not price alone.
demand_context	event/weather/holiday_id, location, valid_from, valid_to, type, severity, source, confidence	Source and confidence required.
pricing_constraint	property_id, room_class_id, effective_from/to, min_rate, max_rate, max_step, excluded_dates	Audit every change to constraints.
forecast_run	run_id, model_version, created_at, data_cutoff_at, horizon, metrics	Every result must be reproducible.
recommendation	recommendation_id, run_id, property/date/room_class, current_rate, suggested_rate, confidence, reason_codes	Immutable output for audit.
recommendation_action	recommendation_id, action, approved_rate, actor, timestamp, reason	Record approved, edited, rejected and published decisions.
actual_outcome	property/date/room_class, realized_occupancy, ADR, RevPAR, room_revenue, no_shows, cancellations	Used for evaluation and monitoring.

9.2 CSV-first integration plan

The client has confirmed CSV export availability. Therefore, the MVP should begin with controlled CSV imports rather than delaying the pilot for vendor API access. Each property gets its own mapping profile because the two franchise systems will not necessarily use the same column names, status codes, date formats or room-type codes.

Export requested	Minimum fields	Frequency
Future reservations / on-books	Reservation ID (pseudonymous), created/updated timestamp, arrival, departure, room type, status, nightly rate, channel/rate plan	Daily; ideally delta updates more frequently
Historical reservations	Same fields plus actual nightly revenue, cancellation/no-show status and tax/fee breakdown	One-time initial load plus incremental daily load
Daily inventory / occupancy	Stay date, room type, physical/sellable inventory, OTB rooms, available rooms, out-of-order rooms	Daily snapshot; preserve capture timestamp
Current rates / future rate calendar	Stay date, room type, current/base rate, rate plan, restrictions, source channel where available	Daily and after major rate change
Night audit / revenue summary	Room revenue ex tax, taxes, fees, payments/tenders, refunds/adjustments, ledger totals	Daily
Room/property configuration	Room classes, capacity, amenities, brand standards, min/max pricing rules	On change
Competitive set / RateGain sample	Competitor, stay date, price, currency, room category, availability, cancellation/refund terms, taxes/fees, timestamp	Daily or agreed licensed schedule

9.3 API integration options

- SynXis: Sabre's public SynXis Property Hub Reservation API documentation shows reservation retrieval and reservation detail fields including stay dates, room types, nightly amounts, taxes and fees. It requires authentication and property/vendor identifiers. This does not prove this client has API access; confirm vendor onboarding and franchise authorization. [R3]
- ChoiceADVANTAGE: the client has confirmed CSV availability. No direct Choice API access is assumed in this specification; request written API/export guidance from Choice or the property's authorized provider.

- RateGain: its RMS partner page describes REST APIs for hotel search, shopping schedule configuration, rate-change updates to an RMS and historical rate evolution. Commercial terms, eligibility, property coverage and right to embed data must be agreed. [R4]

9.4 Data quality and reconciliation rules

- Every import must record property, source, upload time, business date, file hash, schema version, row counts and validation result.
- Deduplicate reservation updates using source reservation ID plus last-modified timestamp; preserve change history.
- Normalize all stay dates in the property timezone (America/Chicago for Fargo), while storing UTC timestamps for ingestion and events.
- Validate date ranges, currency, non-negative room counts, valid rate ranges, status values and room-class mapping.
- Reconcile nightly room revenue, tax, fees and payment totals with the source reports. Differences must be visible, not silently corrected.
- Keep raw CSVs and normalized records separately so mapping defects can be corrected and reprocessed.
- Flag missing snapshot dates; never backfill booking pace with guessed historical on-books counts.

10. Dashboard, reporting and AI experience

10.1 Pricing workbench

Screen	Required content
Portfolio overview	Both properties; current occupancy; future OTB; ADR; RevPAR; revenue; data freshness; high-priority alerts.
Pricing calendar	Rows for stay dates and room classes; current rate; Exora recommendation; rate difference; OTB occupancy; forecast occupancy; comp-set index; confidence; approval status.
Recommendation detail	Top drivers; demand forecast; competitor sample; data source timestamps; floors/ceilings; candidate-rate scenario comparison; reason codes.
Competitive set	Competitor selection, room/terms matching, current and historical rates, availability and collection time.
Approvals	Approve, edit or reject; mandatory reason for material override; user/timestamp; export approved rate sheet.
Performance	Forecast vs actual occupancy; ADR/RevPAR; revenue; pickup; recommendations accepted; overrides; model error.

10.2 Operations and financial intelligence module

- Daily briefing: previous-day performance, key variances, missing reports, significant incidents and action items.
- Financial checks: compare room revenue, taxes, payments/tenders, refunds and ledger totals using deterministic reconciliation rules.
- Revenue metrics: ADR, RevPAR, occupancy, room revenue excluding tax, revenue trends and channel mix.
- Reservation monitoring: cancellations, no-shows, booking pace changes and unusual patterns.
- Operations signals: rooms out of order, maintenance delays, unclosed shift issues and staff-note classification where reports contain those fields.

10.3 AI assistant boundaries

Use LLM for	Do not use LLM for
Summarize staff notes and daily reports; classify incident narratives; explain metrics; answer natural-language questions over curated aggregates; draft management briefings.	Authoritative arithmetic; ledger reconciliation; deciding final prices; directly changing PMS rates; inventing missing data; claiming cause from correlation; reading raw guest PII unnecessarily.

11. Security, governance and contractual controls

CRITICAL REVQI CONTRACT CHECK

The 2025 La Quinta RevIQ supplement includes restrictions on sharing RevIQ outputs with third parties, using RevIQ to process third-party data or as a service provider, and using the system to develop a commercial offering directly or

indirectly competing with IDeaS. This is not confirmed to be the Travelodge client's executed contract, and it is not a legal opinion. Obtain and review the actual Travelodge franchise/technology agreement before Exora receives, stores, exports, benchmarks, shares or trains on any RevIQ output. Build independently from authorized PMS/source data and licensed/public data unless written permission clearly allows more. [R1]

11.1 Access and privacy

- Use property-level service accounts and role-based access; separate Travelodge and Quality Suites data by tenant/property permissions.
- Collect the minimum data needed for revenue analytics. Prefer pseudonymous reservation IDs; exclude guest names, email, phone, payment-card data and identity documents from the pricing dataset.
- Never ingest payment card numbers or security codes. Financial reporting should use totals, tender types and transaction references where approved, not cardholder data.
- Encrypt data in transit and at rest; store credentials/secrets in a secret manager; rotate API keys and maintain an access audit log.
- Apply retention and deletion schedules to raw reports, reservation records and external market data consistent with client contracts and vendor licensing.
- Use a data-processing agreement covering data ownership, purposes, sub-processors, retention, breach notice and derived data/model rights.

11.2 Pricing governance

- Each property maintains its own floor, ceiling, room-class relationships, blocked dates, rate-change limits and authorized approvers.
- The recommendation record stores model version, feature/data cutoff, inputs summary, constraints applied and reason codes.
- Manager approval is required for every live rate change in the pilot. No autonomous write-back until the owner, franchise/vendor and Exora security reviews approve it.
- Provide rollback to the last approved rate and an exportable audit history.
- Avoid personalized pricing based on individual guest characteristics. Keep the decision at property/date/room-class level.

12. Testing, backtesting and success measures

12.1 Data acceptance checks

Check	Proposed acceptance criterion
Reservation coverage	All provided source rows either load or appear in an exception report; no silent drops.
Deduplication	Repeated import of same file produces no duplicate reservation-night rows.
Financial reconciliation	Room revenue/tax/fee totals reconcile to agreed source reports within documented tolerance; all differences are explained.
Inventory consistency	OTB + available + out-of-order reconciles to physical inventory under the client's definitions.
Snapshot continuity	Daily snapshots retained for the chosen future horizon; missing days visible.
Mapping traceability	Every canonical field maps to source file/column and transformation version.

12.2 Forecast/model evaluation

- Use rolling-origin backtesting: train only on information available at each historical cutoff and forecast future stay dates.
- Compare forecasts with simple baselines (same weekday/season historical average, seasonal-naive and current on-books trajectory).
- Track occupancy forecast MAE/WAPE or another agreed error metric by property and horizon; do not report one aggregate score without property/horizon breakdown.
- Evaluate recommendation outcomes by accepted vs rejected vs edited, but do not interpret acceptance as proof of uplift.
- Price optimization needs a credible estimate of demand response to price. Use controlled, manager-approved tests or defensible quasi-experimental methods; do not infer elasticity from correlation alone.

12.3 Business KPIs

KPI	Purpose
ADR	Average realized room price excluding tax, with agreed definition.
Occupancy	Utilization of sellable inventory.
RevPAR	Revenue per available sellable room; core revenue-yield metric.
Net room revenue	Revenue after known channel/distribution costs where available.
Forecast accuracy	Whether demand/occupancy estimates are reliable enough for pricing decisions.
Recommendation adoption	Share approved, edited and rejected; capture reasons.
Rate-change turnaround	Time from recommendation to manager decision / publication.
Manual reporting time	Manager time spent preparing/reviewing reports before vs after deployment.
Exception resolution time	Time to investigate and resolve discrepancies/operational alerts.

NO PREMATURE ROI CLAIMS

Do not promise a fixed revenue uplift before historical backtesting and a controlled pilot. Report measured results with the observation period, comparator and limitations. The initial pilot proves data quality, forecast usefulness and manager adoption—not guaranteed incremental revenue.

13. Implementation roadmap and acceptance criteria

Phase	Work	Exit gate / deliverable
0. Discovery & permissions	Collect executed agreement excerpts, CSV samples, room configuration, current workflows, rate guardrails, revIQ/Choice usage details and vendor access permissions.	Signed-off data dictionary, source inventory, legal/access checklist and agreed KPI definitions.
1. Data foundation	Build CSV mapping profiles; raw archive; normalization; canonical reservation-night, inventory and rate snapshots; data-quality reports.	Both properties load repeatably; core metrics reconcile; snapshot process running daily.
2. Baseline pricing prototype	Implement comparable-date baseline and transparent rules; 30-day rate calendar; floors/ceilings; manager explanations.	Manager can review recommendations based on actual property data; no live publishing.
3. Forecasting & market context	Add rolling forecasts, booking pace, competitor-rate feed/sample, event/weather signals, confidence scoring.	Backtest vs baselines; validate competitive-set comparability and provider economics.
4. Pricing workbench & approvals	Build recommendation detail, approve/edit/reject, audit history, exports, property-separated permissions.	Manager workflow accepted; every decision traceable.
5. Pilot & calibration	Run shadow mode, compare recommendations with existing decisions and realized outcomes; adjust guardrails.	Documented pilot report, error metrics, usability sign-off and go/no-go decision.
6. Optional write-back & expansion	Obtain approvals, implement authorized PMS/CRS rate publishing, rollback and expand additional properties.	Vendor certification and signed customer approval; production monitoring and rollback tested.
7. Operations intelligence	Add night-report ingestion, financial reconciliation, anomaly rules, daily briefings and AI assistant.	Reports correctly parsed and reconciled; AI narrative always references validated metrics/source.

13.1 MVP acceptance criteria

- Both properties can be represented separately with their franchise, source-system and room-class mappings.
- Authorized CSV exports can be imported repeatedly with idempotent processing and an audit record.
- The system creates daily future stay-date snapshots and computes OTB occupancy and pickup.
- The dashboard shows a 30-day rate calendar with current rate, suggested rate, projected occupancy, source freshness, confidence and explanation.
- Each suggestion respects property-specific floors, ceilings and maximum rate-change limits.

- Competitor comparisons are only made when stay date, room class, availability and booking terms are sufficiently matched; otherwise they are marked non-comparable.
- Managers can approve, edit or reject each suggestion; all actions and override reasons are stored.
- No rate is written to a live property system during the MVP without explicit authorization and approval.
- Backtesting uses time-correct historical cutoffs and reports metrics against a documented baseline.
- All calculations can be traced to source records and metric definitions.

14. Risks, assumptions and discovery checklist

Risk / open item	Why it matters	Mitigation / action
Exact franchise system modules unknown	"SynXis" may refer to a specific PMS/CRS/property hub configuration; different APIs and data rights apply.	Confirm exact product/version and authorized exports for Travelodge.
ChoiceADVANTAGE data granularity unknown	Public description is high-level; property-level fields may not expose all model inputs.	Inspect actual sample CSVs; ask Choice/provider for export and API catalogue.
RevIQ product tier/configuration unknown	Standard vs Premium and property-specific features affect comparison.	Ask manager for current screenshots and subscription/tier; do not access restricted output until contract review.
Historical snapshots absent	Current bookings do not reconstruct old booking pace.	Start daily snapshots immediately; obtain historical dated exports if available.
Small data volume	Two properties and ~120 rooms may be insufficient for a highly complex model.	Pool data carefully with property effects; use simple baselines; validate model stability.
RateGain price / coverage unknown	External data may cost more than the value for two properties or miss local competitors.	Request a Fargo sample, commercial quote, usage rights and embedded/OEM terms before purchase.
Competitor price comparability	Different room classes, refund terms, fees, tax and availability create false comparisons.	Normalize, store source metadata and suppress weak matches.
Contract / use restrictions	RevIQ terms may restrict sharing outputs and competitive use; property system terms may restrict integrations.	Legal review of the client's signed agreements and written third-party authorization.
Automated price publishing	Wrong price can affect many bookings and franchise operations.	Manual approval first; separate certification, rate limits and rollback.
Revenue definitions mismatch	ADR/RevPAR/tax/fees can differ across systems.	Agree calculation standards and reconcile before modelling.

14.1 Client discovery checklist

- Confirm exact property legal/operator names, brand names, street addresses, room counts and room classes for both motels.
- Confirm whether RevIQ is used for Travelodge only or for both properties; obtain tier/configuration and the actual Travelodge contract terms.
- Confirm whether Quality Suites actively uses the built-in choiceADVANTAGE rate/selling management tool and whether any separate RMS is used.
- Obtain anonymized sample exports: future reservations, historical reservations, current/future rate calendar, daily occupancy/inventory and night audit/ledger summaries for each property.
- Confirm 12–24 months of history availability, daily snapshot history, cancellation/no-show fields, booking timestamps and room-level/class-level detail.
- Obtain the current competitive set and ask how they decide which nearby motels are comparable.
- Confirm rate floor/ceiling, max daily rate change, franchise promotion/parity rules, who can approve and whether the client expects auto-publishing.
- Ask which reports the manager reads daily and define the exact calculations for ADR, revenue excluding tax, cash flow and ledger discrepancies.
- Obtain RateGain API sample, Fargo coverage report, commercial quote, rate limits, licensing, retention and white-label/embedded rights.
- Agree pilot success measures, pilot period, budget owner, monthly willingness-to-pay after demo and sign-off authority.

15. References and source notes

The following public sources informed the documented product capabilities and integration options. Accessed 04 November 2026. These sources do not replace the client's executed franchise, technology or vendor agreements.

[R1] 2025 La Quinta Franchise Disclosure Document — RevIQ Standard/Premium supplement (Q1 2025; feature description and use restrictions). https://pub-22da695f2b6542a48e234bc59bc67d23.r2.dev/fdds/2025/LA_QUINTA_FRANCHISING_LLC_cmfed9a2002rf1vbtlvcf87l3.pdf *Applies to La Quinta source document; not confirmed as the Travelodge agreement. See the RevIQ supplement and system-use restrictions.*

[R2] Choice Hotels International, Inc. 2025 Form 10-K (filed 2026). https://www.sec.gov/Archives/edgar/data/1046311/000104631126000008/chh-20251231.htm *Describes choiceADVANTAGE room inventory/rates/reservations and revenue-management suggestions based on past performance and projected occupancy.*

[R3] Sabre SynXis Property Hub — Reservation API documentation. https://developer.synxis.com/pms/property_hub/reservation *Shows reservation retrieval and related reservation/room operations; authenticated access and appropriate property/vendor identifiers are required.*

[R4] RateGain — Solutions for RMS Companies / Rate Intelligence APIs. https://rategain.com/solutions-for-rms-companies/ *Describes REST-based RMS integration, rate shopping schedules, historical rate evolution and rate-change feeds. Commercial terms still need validation.*

[R5] National Weather Service — API documentation. https://www.weather.gov/documentation/services-web-API *Public API for forecasts, alerts and observations; published as open data subject to reasonable usage limits.*

[R6] Ticketmaster Developer Portal — Discovery API. https://developer.ticketmaster.com/products-and-docs/apis/discovery-manual/v2/ *Event search API; coverage is not a complete calendar of every local Fargo event.*

[R7] Choice Hotels 2023 Form 10-K (historical disclosure). https://s201.q4cdn.com/538915302/files/doc_financials/2023/ar/chh-choice-hotels-international-inc-10-k-2023.pdf *Earlier description of choiceADVANTAGE's proprietary rate-suggestion feature; the 2025 Form 10-K is preferred for current description.*

15.1 Evidence labels used in this specification

- Client-reported: stated in the WhatsApp conversation or prior discovery notes (for example, CSV exports are available and managers review daily/night reports).
- Publicly documented: stated in a linked vendor filing, official API reference or official product page.
- Proposed / inferred: Exora design recommendation, not a claim about an incumbent vendor's internal implementation.
- To confirm: requires a client export, contract, vendor response or technical test before implementation.

FINAL BUILD RECOMMENDATION

Start with a CSV-first, independently sourced data foundation. Implement a transparent 30-day recommendation calendar with a common historical-performance / projected-occupancy baseline, then add room-class, season/event, competitor, booking-pace, channel-cost and weather features as data quality and commercial rights permit. Keep the engine advisory until a backtested pilot is accepted and all live rate-publishing permissions are documented.